

Bridging Cross-View Retrieval with GPS: A GeoCLIP + Sample4Geo Hybrid for Landmark Geolocalisation

“Can we use GeoCLIP’s GPS guess to narrow Sample4Geo’s satellite search and improve cross-view retrieval?”

Mateusz Zbyslaw · Edvin Smajlovic · Konstantinos Papadopoulos · Nikolaos Iliakis Supervisor - Oskar Kristoffersen

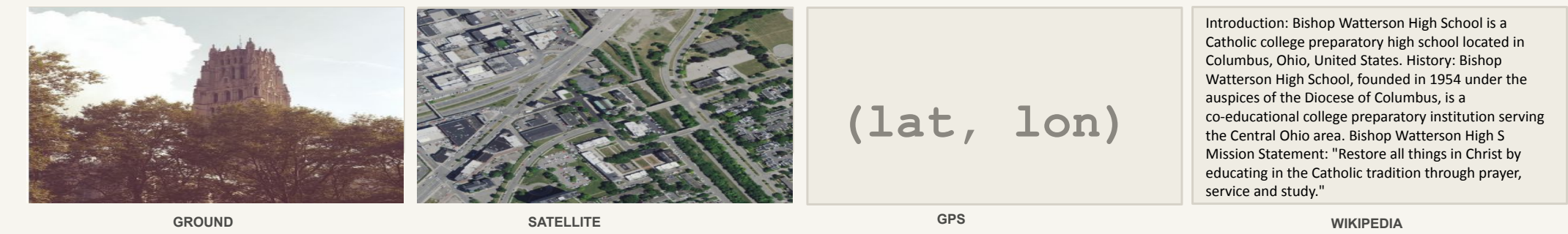
SECTION 01 · PROBLEM & DATASET

Can use additional modalities to improve the performance of localization tasks?

The dataset we use for this project is the **MMLandmarks**: An open-source US-landmark benchmark with 1-to-1 ground/satellite correspondence + per-landmark Wikipedia text and GPS

17,557 train landmarks1,000 query landmarks100k satellite gallery4 modalities · ground · satellite · GPS · text

FIG 1 · FOUR MODALITIES, ONE LANDMARK



SECTION 02 · CROSS-VIEW RETRIEVAL & SAMPLE4GEO

Matching ground view images to satellite images

Cross-view retrieval requires matching a ground-level photo to the correct satellite image despite a large viewpoint gap - a task where standard visual features fail. We fine-tune a single shared ConvNeXt-Base encoder with symmetric InfoNCE and a two-phase hard-negative schedule that transitions from GPS-proximity sampling to model-guided similarity sampling.

- **Single shared encoder** - one ConvNeXt-Base (88M params) for both modalities; shared weights
- **Symmetric InfoNCE** - $L = \frac{1}{2}(L_{g \rightarrow s} + L_{s \rightarrow g})$; all batch images act as negatives, learnable temperature τ
- **Multi-positive loss (v3/v4)** - $K=2$ ground images per landmark per step
- **Hard negatives** - GPS-Sampling (epochs 1-4) $\rightarrow$ Dynamic Similarity Sampling (DSS) (epoch > 4)
- **LR schedule** - AdamW, lr=1e-4; 3-epoch linear warmup from 1e-5, cosine decay to 1e-6

FIG 3 · Sampling strategies

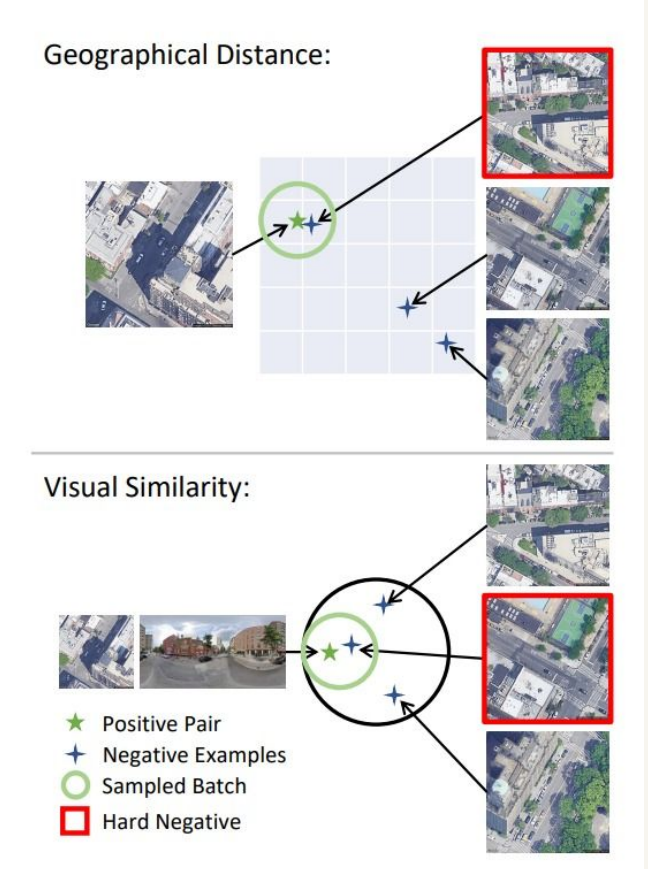
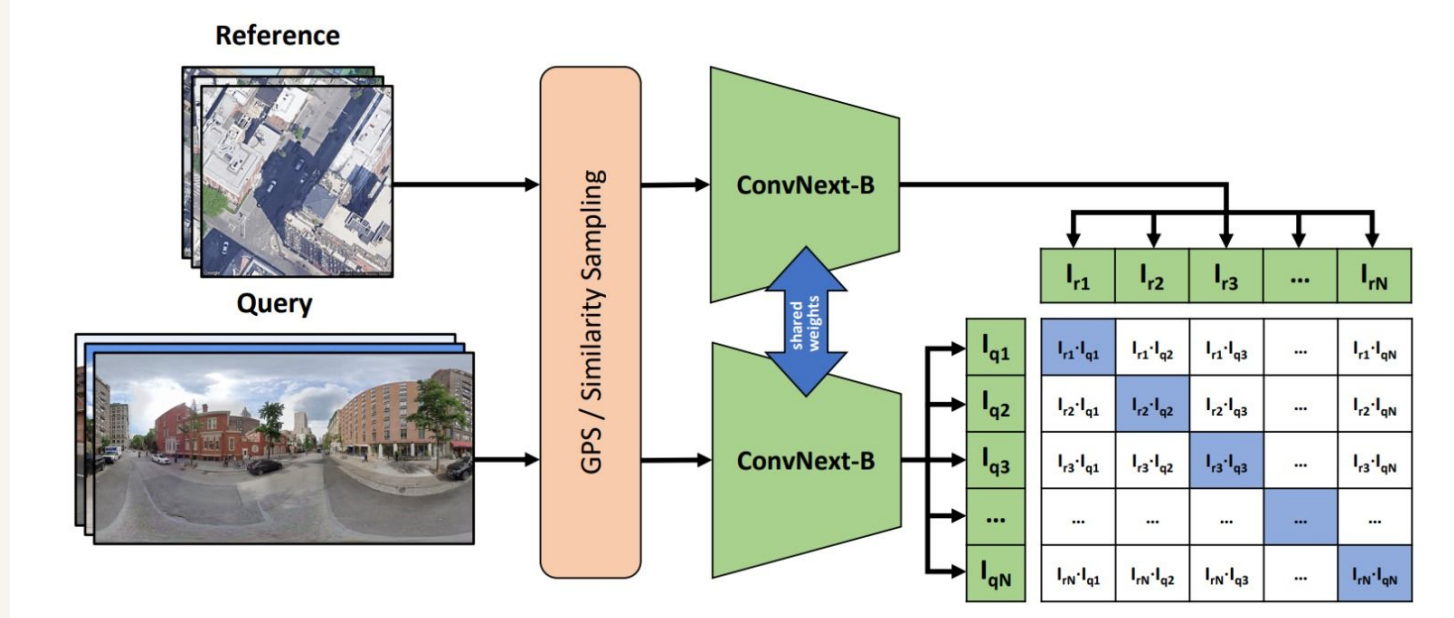


FIG 2 · Architecture overview



SECTION 09 · Final Results

Multi Image Results

Sample4Geo	R@1	R@10	R@25	MAP@1K
v2 - Weighted-mean	17.5%	42.1%	51.0%	25.2%
Hybrid	R@1	R@10	R@25	MAP@1K
Transformer GeoCLIP + Weighted-mean S4G	23.30%	33.30%	40.60%	23.70%

SECTION 10 · Statistics

Recall, MAP, and Per landmark evaluation

- **Recall@K**
- **MAP@K**
- **Per landmark evaluation**

Since our single image models produce N predictions per landmark and our multi-image models produce just one, all statistics in our poster are measured per landmark. This has the added benefit of weighing landmarks with more images equally with landmarks with less images.

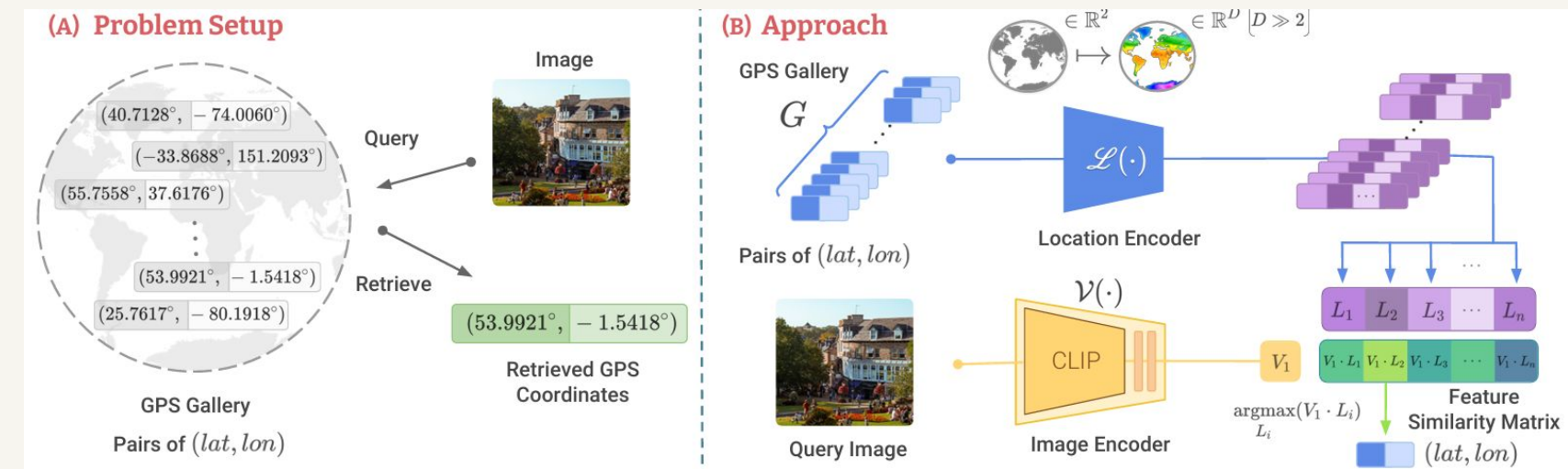
SECTION 03 · GEOCLIP AS A BRIDGE

Predicting GPS coordinates from the ground truth

GeoCLIP turns a single ground photo into a GPS coordinate

Frozen CLIP ViT-L image encoder + small linear head; a separate **location encoder** (Equal-Earth $\rightarrow$ Random Fourier Features $\rightarrow$ hierarchical MLP) embeds GPS.

Both produce 512-d vectors trained with a contrastive loss $\rightarrow$ at inference: $\text{argmax}_{\text{GPS}} \cos(\text{img}, \text{GPS})$



SECTION 04 · HYBRID PIPELINE

Our proposal: A hybrid model for cross-view retrieval.

FIG 3 · TWO-STAGE PIPELINE



SECTION 05 · Preliminary Results

Single Image Results

Sample4Geo	R@1	R@10	R@25	MAP@1K
Zero-shot - No MMLandmarks training - pure ImageNet features, ConvNeXt-Base 384px	0.16%	1.25%	2.36%	0.55%
v2 - Batch 64 (63 negatives/step), single ground image per landmark, standard InfoNCE	6.45%	22.29%	30.92%	11.50%
v3 - 384px + stronger backbone, batch 16 with AMP, multi-image InfoNCE (K=2), label smoothing	6.95%	17.45%	22.29%	10.57%
v4 - Batch 64, multi-image InfoNCE (K=2) + label smoothing - same backbone/res as v2 (ablation)	6.71%	20.53%	28.26%	11.31%
Hybrid	R@1	R@10	R@25	MAP@1K
Finetuned Hybrid	9.1%	15.90%	20.60%	22.10%

SECTION 06 · Paper vs Index

Making sure the predictions generalize

Should the candidate GPS or satellite images be included the ground truth?

Prediction vs Retrieval

SECTION 07 · GEOCLIP IS AN UPPER-BOUND

Geoclip upper-bounding our hybrid model

If GeoCLIP’s GPS error exceeds the filter radius, the correct satellite is dropped before Sample4Geo ever sees it

Fine-tuning vanilla GeoCLIP on MMLandmarks hardly improved performance

Hypothesis. MMLandmarks contains a varying amount of ground images per landmark. Are we able to utilize this to increase Geoclips performance?

Multi-image Geoclip

Each landmark has 1 – ~150 ground photos (avg 18). Need to aggregate K embeddings $\rightarrow$ one landmark embedding.

Weighted-mean. Mean of K embeds, then re-weight each by cosine-sim to that mean (soft-max). Emphasizes agreement

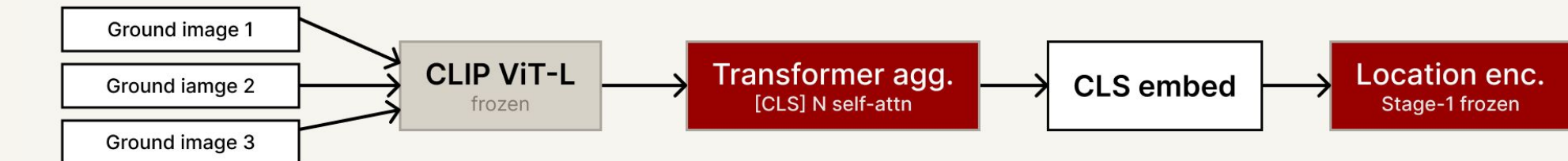
Transformer (attn). Encoder-only transformer over N image embeds + [CLS] token $\rightarrow$ one landmark embed.

SECTION 08 · NEW MULTI-IMAGE GEOCLIP

Transformer Geoclip

Stage 1 - Freeze Geoclip while training the transformer normally. **Stage 2** - Unfreeze Geoclip and lower the learning rate, to finetune the pipeline

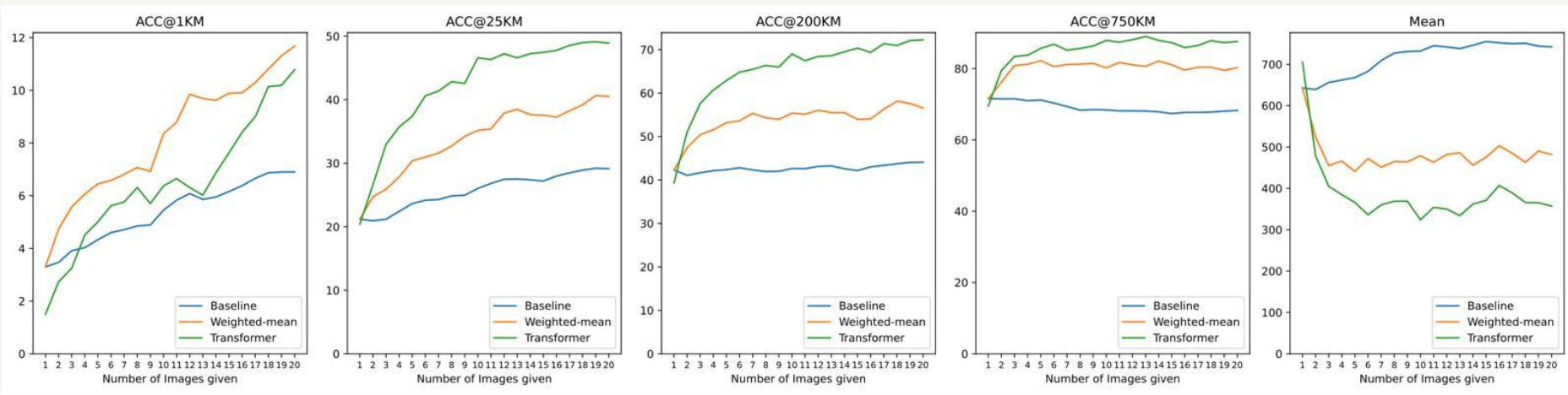
Goal is for the model to see each image in the context of the other images when creating its prediction embedding



Results:

Performance per image given to the model

Transformer and weighted mean both outperform Geoclip in both accuracy and uncertainty with more images



SECTION 11 · Qualitative Results

Qualitative comparison of v3 Sample4Geo, Finetuned Hybrid, and Multi-image Hybrid



SECTION 12 · Conclusion

Conclusion and lessons

- **Domain training for Sample4Geo.** Training Sample4Geo on our dataset made a massive improvement compared to zeroshot. This makes sense due to how different the dataset it was trained on is.
- **CLIP backbone dominated Geoclip.** In our tests, fine-tuning GeoCLIP on MMLandmarks barely improved performance. With 17k US landmarks against a backbone trained on 4M+ global photos, the CLIP image encoder is robust.
- **Fair evaluation.** Per-image statistics were originally computed, but didn’t let us compare single-image models to multi-images ones. Making sure all the statistics were computed the same and fair took a lot of time
- **Uncertainty Quantification.** Aggregating the K ground photos of each landmark into one embedding (transformer with [CLS]) cut mean GPS error from 650 km to 400km
- **Our ideas worked.** Our single-image hybrid model saw a 50% improvement over our finetuned Sample4Geo, and our multi-image tests showed similar good improvement for adding more images.

SECTION 13 · Use of AI

AI Disclaimer

- We used Claude to help write and debug code for training, evaluation, data processing and to summarize code branches to keep the team updated.
- We used Claude and ChatGPT to discuss ideas, understand papers, check mathematical reasoning and to help organize and improve the wording of the poster.
- All experiments, model decisions, results, and conclusions were made and checked by our team.
- Any AI-generated content was reviewed, edited, and verified before being used.